IKPM Portal: Final Workflow Gap & Alignment Report

Date: 31 October 2026 **Scope:** Branch `10/30/2026` including uncommitted changes (8 files), checked against *IKPM CRM Requirements & Developer Build Guide v1.1*, plus your field model (below). **Build health:** TypeScript type-check passes; 263/263 unit tests pass. Firestore rules emulator tests were not run (they need the emulator).

0. Your field model (what the system must follow)

Person	Has portal login?	What they do in the CRM
Operations Manager (`operations`)	Yes	Creates contracts, schedules visits, assigns a **Team Leader** to each visit
Team Leader / Coordinator (`team_leader`)	Yes	Coordinates their PCOs in person / WhatsApp, **submits the job report**, ticks which PCOs did the work
PCO / Technician	**No**	Exists only as an HR **Employee** record (`EMP-xxxx`) with a Team Leader set

Rule this creates: two kinds of IDs, never mixed:

- **Login IDs** (Team Leader) → who can *see and submit* the job → `assignedTeamLeaderId`, `assignedTechnicianIds`
- **Employee IDs** (PCOs) → who *did the work* (payroll, history, complaints) → `assignedPcoEmployeeIds` (planned), `performedByPcoIds` (actual)

Almost every break below happens because the code mixes these two today.

1. Critical: the issue you reported (Team Leader can't pick their own PCOs, or more than one)

#	Where	Problem	Effect
1.1	Report form, `JobsListPage.tsx` L1267-1277	"PCO who performed" lists **login users with role technician**. PCOs have no login, so the list is **empty** (or shows wrong people)	Team Leader cannot record who did the job
1.2	Same	Single-select dropdown	Cannot record 2-3 PCOs on one job
1.3	Same	No filter by Team Leader	Team A leader would see Team B/C people
1.4	Submit handler, L475-504	Saves a login ID into `performedByPcoIds`; payroll counts by **Employee ID** (`payroll.ts` L47-48)	**PCOs get 0 verified jobs → wrong pay / incentive / score**
1.5	Submit handler, L501	Overwrites `assignedTechnicianIds` with `[leadPcoId, pco]`	Can drop the Team Leader / other assignees' access
1.6	Schedule generator (uncommitted), L47-100	Puts **Employee IDs** into `assignedTechnicianIds` (a login field) and also lists *all* PCOs	Meaningless login entries; "same team" conflict filter doesn't filter
1.7	Schedule generator	Detects Team Leaders by checking label text contains "(Team Leader)"	Breaks if the label changes
1.8	`dataService.ts` L1913-1919 (uncommitted)	Double-booking used to block; now only `console.warn`	A Team Leader can be double-booked silently

#	Where	Problem	Effect
1.9	New-job form L896-915	Field says "Assign Team Member", placeholder says "Select a team leader…", list mixes Ops Managers + TLs + technicians	Confusing; wrong person can be picked
1.10	`dataService.ts` L450-456	A Team Leader with "own"/"team" data scope only loads jobs they own or are in `assignedTechnicianIds` for. Jobs with only `assignedTeamLeaderId` set don't load	TL may not see jobs assigned to them

Fix

1. **One helper** `getTeamPcos(tlUid)`: active HR employees with `role = 'pco'` and `reportingCoordinatorId = tlUid`. Every PCO picker uses only this.
2. **Scheduling:** Team Leader is required (only `team_leader` users listed). PCOs are optional tick-boxes from that TL's team → saved to `assignedPcoEmployeeIds`. `assignedTechnicianIds = [tlUid]` only.
3. **Report form, "Who performed this visit":** tick-boxes of the TL's PCOs, planned ones pre-ticked, minimum 1 (or "I did it myself"), plus "+ Add PCO from another team (reason)". Save Employee IDs to `performedByPcoIds`; **do not touch** `assignedTechnicianIds`.
4. **Double-booking:** block on the Team Leader's slot again; PCO overlap shows a warning only.
5. **Job loading:** add a query on `assignedTeamLeaderId == uid` so TLs always see their jobs.
6. **Job card:** "Team A · Imran Hussain (TL) · Usman, Bilal".
7. **Payroll test:** one job with 3 PCOs → each gets +1 verified job.

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2. High: other workflow breaks

#	Problem	Where	Fix
2.1	QC "Rejected" / "Needs revisit" still sets **verified = true** → invoice unlocked	`JobsListPage.tsx` L378-390	Two buttons: **Verify** / **Send back (reason)**. Send back returns job to TL
2.2	QC form pre-filled ("Muhammad Asif", "…confirmed compliant…")	L180-187	Supervisor = logged-in user; notes empty
2.3	Challan photo mandatory for **every** outcome, but "Client blocked access" has no signed challan	L451-459	Challan + sign-off required only for *Completed / Partial*
2.4	"Postponed" leaves job *In Progress* forever	L493	Postponed → opens Reschedule; Client blocked → cancel with reason code
2.5	Every manual job gets a **fake chemical** (Premise 200 SC, 1 L)	L344-356	Start with empty chemical list
2.6	Late flag uses `> 1` day; SRS says same day	L491	`> 0`
2.7	Manual job ID = `jobs.length + 1` (can repeat); site always `SITE-01`	L317, L325	Use the ID counter; pick real site/branch
2.8	No start gate: pre-check (uniform, safety gear, chemical issue) and **before photos** not required (SRS §5.5, §18)	`workflow.ts` L290-301	TL confirms pre-check + uploads before photos on Start
2.9	No **chemical-match** step at QC (SRS §5.5)	QC modal	QC re-enters chemical from signed report; mismatch flags & asks reason

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3. High: security regressions (uncommitted `firestore.rules`): fix before deploy

#	Rule	Problem	SRS
3.1	`service_jobs` L341, `contracts` L323	`allow read: if active()`: HR, Finance, anyone reads all jobs & contract values	§3.1
3.2	`service_jobs`	Anyone with `pco: edit` (incl. CRO `sales`) can create/update any job and change assignees; the old protective clause is now bypassed by a broader OR	§3.1
3.3	`contracts`	`sales`, `team_leader` can update **any** contract field	§3.1
3.4	`employee_private` L530	Uses `canEdit('hr')`; your changes gave Ops Manager & Admin `hr: edit` → they can read **CNIC / bank**	§3.1: HR & CEO only
3.5	`employees` L524	Anyone with `canEdit('jobs')` can edit employee records	§3.1
3.6	`payroll` L551	Extra `active()` clause: every user reads payroll (already committed)	§3.1

Fix: revert to scoped reads; jobs readable by job roles + `assignedTeamLeaderId == uid`; private HR data gated on role `hr`/`super_admin` explicitly (not module edit); remove `active()` from payroll.

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4. Alignment check (do modules agree with each other?)

Area	Misalignment	Fix
Role names	Code: `operations` = Ops Manager, `team_leader` = Coordinator/TL. HR page labels `team_leader` "Team Leader" and `operations_coordinator` "Operations Coordinator" as different things. `staffSeedData.ts` gives the 3 Karachi TLs role `operations` (= Ops Manager)	One label set everywhere: **Operations Manager / Team Leader / PCO**. Fix seed roles
Team definition	Teams exist twice: login users have `teamId` (Team & Access page); PCOs have `reportingCoordinatorId` (HR page)	Team = a Team Leader. PCO's team comes from `reportingCoordinatorId`. Store the TL's **login ID** there (HR page currently allows an Employee ID too)
Attendance	TL sees **all** PCOs company-wide; everyone defaults to *Present 08:30*, so one Save marks the whole company present	TL sees only their team; default = *not marked*; "Mark all present" as an explicit button
Payroll ↔ Jobs	Payroll reads Employee IDs; report saves login IDs	Fixed by §1
Complaints ↔ Jobs	Complaint should show which PCO/TL worked at the client (SRS §5.6); PCO names aren't reliably on jobs today	Fixed by §1 (`performedByPcoIds`); show on client page
Complaint escalation	Status "escalated" exists, but no automatic 24/48 h timer found	Scheduled check using Settings value
Lead → Proposal gate	Could not find the "proposal only after Inspection Complete" barrier (SRS §5.2)	Add gate in proposal builder

Area	Misalignment	Fix
Not built yet (SRS)	Expenses with receipt + senior/Accounts verification; random visits & overdue PCO flag; risk assessment on service jobs	Phase per SRS §9

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5. User-friendly workflow (recommended)

Operations Manager

1. Contract → **Generate schedule** → pick **Team Leader** per branch (one choice) → optional PCOs from that team → Confirm.
2. Dashboard: unassigned visits, team load per day, overdue reports.

Team Leader (main daily user, mobile-first)

"My Team Today" screen: today's jobs + my PCOs + attendance in one place.

Each job is **one screen with 3 steps**, each unlocking the next:

1. **Start**: tick uniform / safety gear / chemicals issued, take before photos.
2. **Service**: chemicals used (pick from list), after photos.
3. **Close-out**: choose outcome:
   • *Completed / Partial* → tick PCOs who worked, challan # + photo, client name & rating → Submit
   • *Client blocked* → reason + photo → job cancelled with code
   • *Postponed* → new date → rescheduled

No separate modals, no fields that don't apply to the chosen outcome.

QC / Ops Manager

Verification queue → open job → re-enter chemical from signed report → **Verify** or **Send back (reason)**. Only verified jobs go to Accounts.

General UX

- Same names for roles everywhere.
- Required fields marked and explained *before* submit (not only an error after).
- Every picker shows only relevant people (my team first).
- Default to empty, not to fake/pre-approved values.

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6. Recommended order

1. Security rules (§3)
2. Team Leader / multi-PCO fix + payroll (§1)
3. QC Verify / Send back, remove fake chemical, late flag (§2.1-2.6)
4. Outcome-aware close-out + start gate (§2.3, 2.4, 2.8)
5. Alignment: role labels, attendance per team, job loading for TL (§4)
6. Chemical match, complaint escalation timer, proposal gate
7. Remaining SRS modules (expenses, random visits)

Note: steps 1-2 edit files that currently have uncommitted changes. Commit or back up first.